Basic Principles of Medicine II
Module: Nervous System & Behavioral Science
Lecture No: NB 09

Neurotransmitter Pathways and Roles

Neuroscience Faculty
St. George’s University

Recommended Reading

Siegel and Sapru, Essential Neuroscience, 4th edition, © Lippincott, Williams & Wilkins (2015), Chapters 6-7.

Online link:

Chapter 6:

https://meded-lwwhealthlibrary-com.sgu.idm.oclc.org/content.aspx?sectionid=196986798&bookid=2482

Chapter 7:

https://meded-lwwhealthlibrary-com.sgu.idm.oclc.org/content.aspx?sectionid=196986923&bookid=2482

Learning Objectives

SOM.MK.I.BPM2.1.NB.1.NSCI.0111	Describe the synthesis/removal, nuclei and projections and receptors tied to acetylcholine
SOM.MK.I.BPM2.1.NB.1.NSCI.0112	Describe the synthesis/removal, nuclei and projections and receptors tied to amino acid transmitters
SOM.MK.I.BPM2.1.NB.1.NSCI.0113	Describe the synthesis/removal, nuclei and projections and receptors tied to catecholamines and serotonin

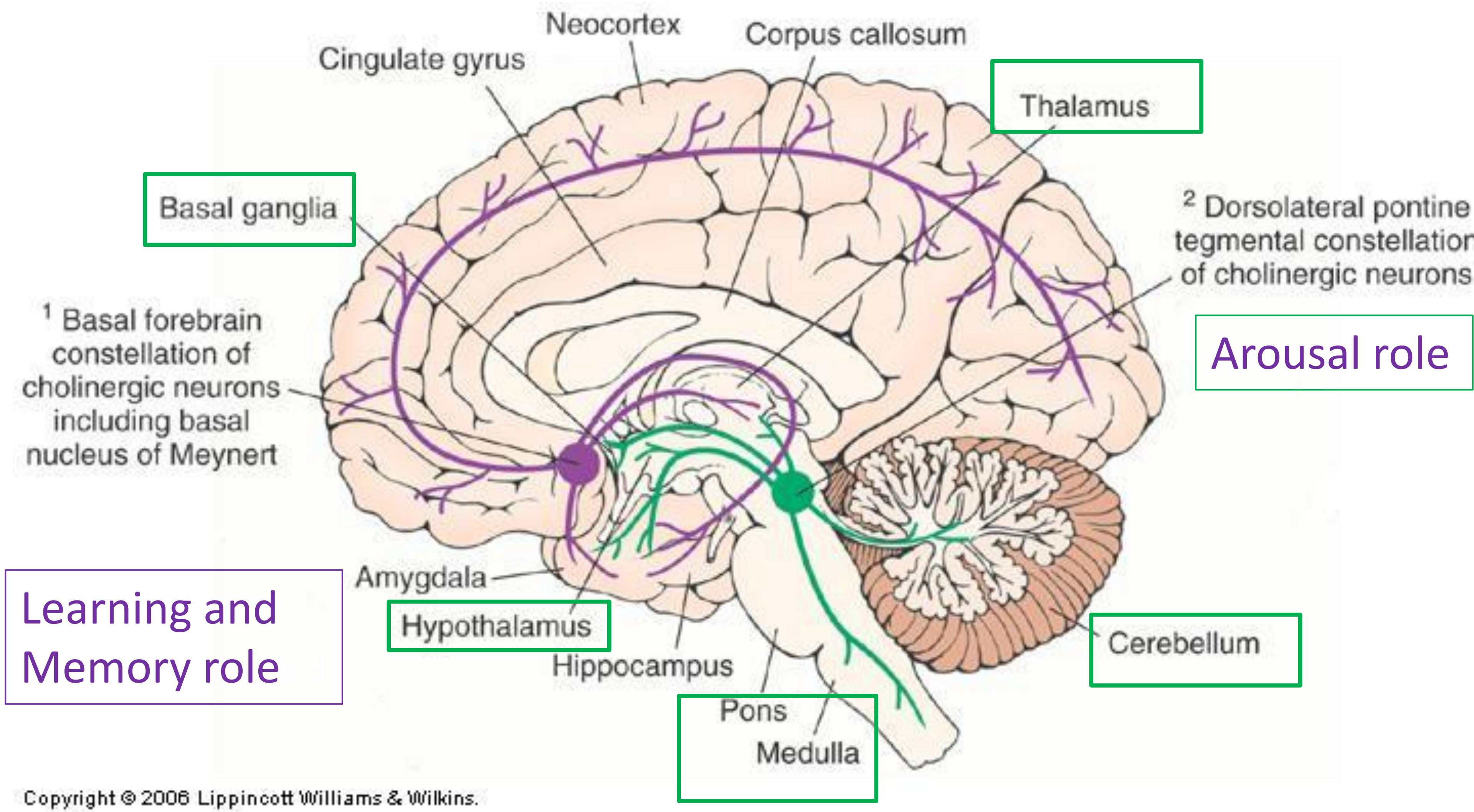
Learning Objectives

- **SOM.MK.I.BPM2.1.NB.1.NSCI.0111** Describe the nuclei and projections and receptors tied to acetylcholine

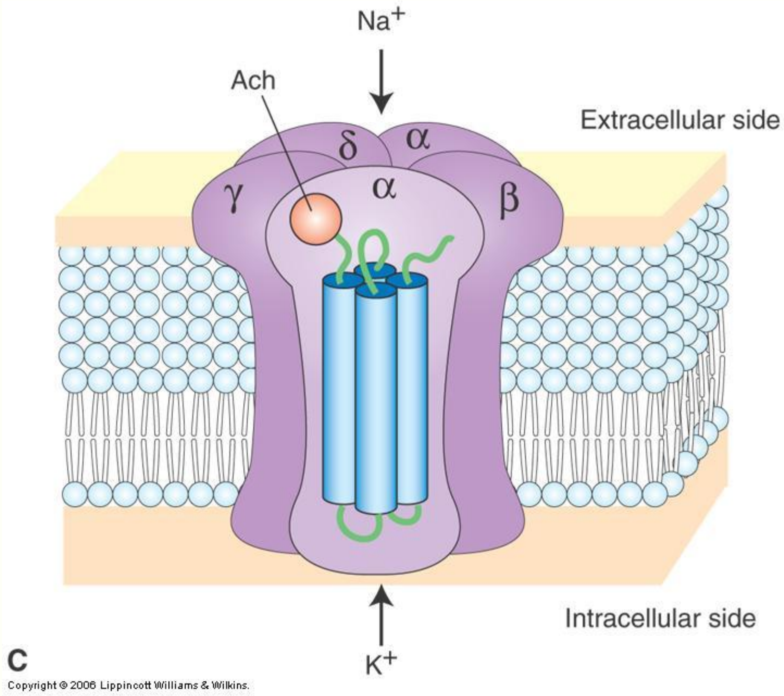
- **SOM.MK.I.BPM2.1.NB.1.NSCI.0112** Describe nuclei and projections and receptors tied to amino acid transmitters

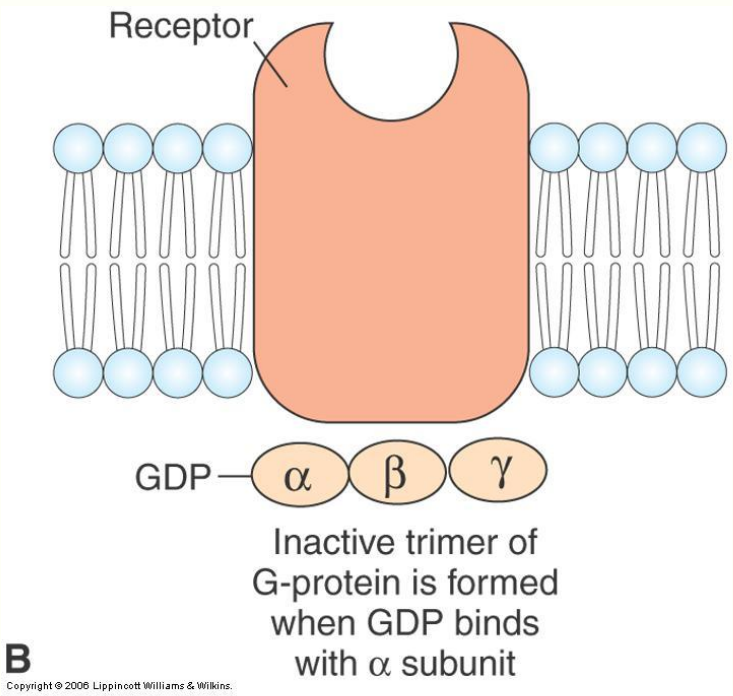
- **SOM.MK.I.BPM2.1.NB.1.NSCI.0113** Describe the nuclei and projections and receptors tied to catecholamines and serotonin

Cholinergic projections: arousal and memory



Ach Receptors: Ionotropic vs Metabotropic





Ionotropic: Nicotinic

Change in ion flux (Na^+, K^+)

Rapid Effects (milliseconds)

Metabotropic: Muscarinic

Metabolic changes, e.g. phosphorylation

Slower effects (seconds to minutes)

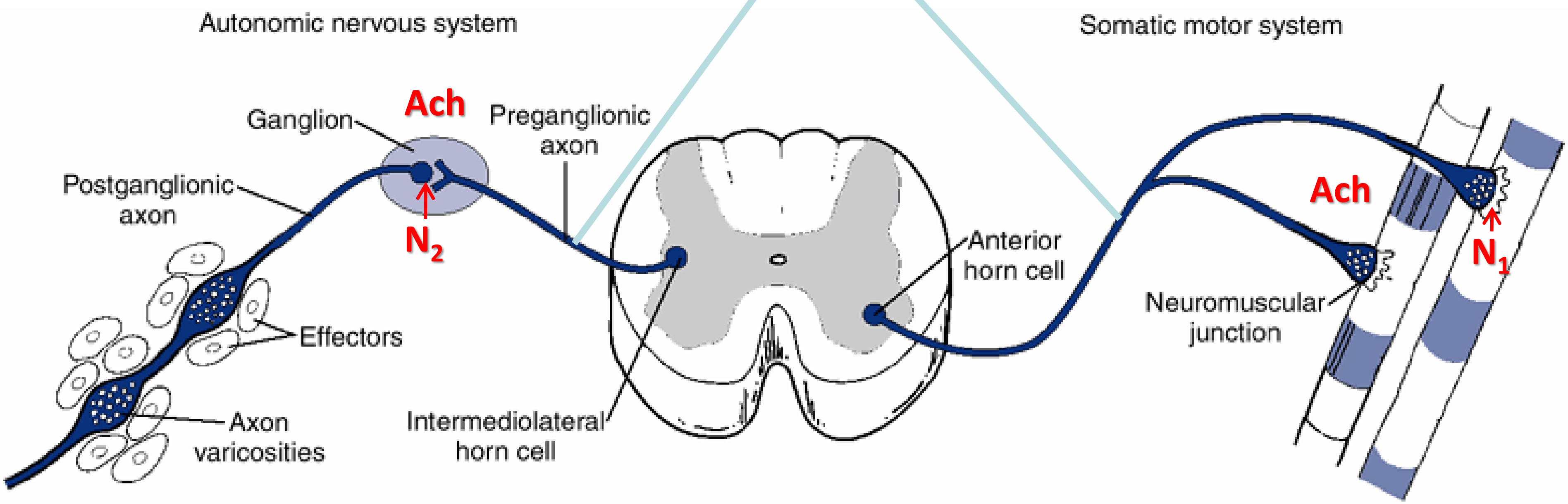
Figs 7.17 and 7.20, Siegel & Sapru.

Central Acetylcholine: Pathology and Therapeutics

- Alzheimer's disease
  - Nucleus basalis of Meynert **degenerates**
  - **AChE inhibitor** approved for treatment
    - Not disease-modifying

Peripheral Cholinergic Neurons

Somatic and preganglionic
efferent fibers are myelinated



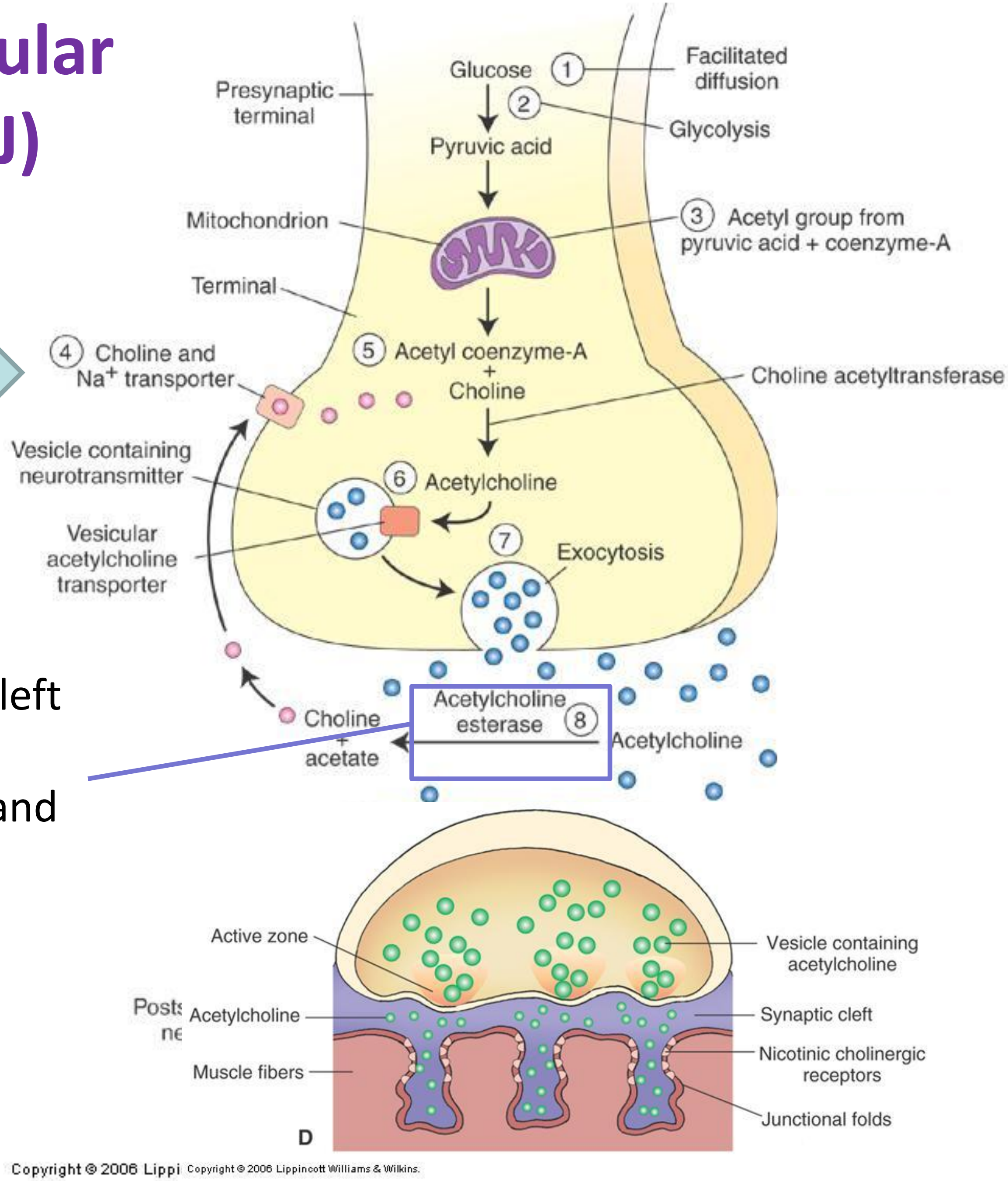
Smooth muscle, cardiac
muscle and glands

Skeletal muscle

The Neuromuscular Junction (NMJ)

Choline/Na symporter

Cholinesterase in synaptic cleft quickly degrades ACh, preventing desensitization and tetany



SOM.MK.I.BPM2.1.NB.1.NSCI.0111

Figures 6.3 and 7.5 from Siegel and Sapru.

Peripheral ACh: Pathology and Therapeutics

- Myasthenia gravis
  - Disrupted cholinergic transmission at neuromuscular junction
  - Treatment
    - AchE (esterase) inhibitors

- SLUD
  - Salivation, lacrimation, urination, defecation
    - **Overactivation** of targets of the cholinergic postganglionic fibers
  - Treatment
    - Muscarinic receptor antagonists
      - Atropine

Lecture Objectives

SOM.MK.I.BPM2.1.NB.1.NSCI.0111	Describe the synthesis/removal, nuclei and projections and receptors tied to acetylcholine
SOM.MK.I.BPM2.1.NB.1.NSCI.0112	Describe the synthesis/removal, nuclei and projections and receptors tied to amino acid transmitters
SOM.MK.I.BPM2.1.NB.1.NSCI.0113	Describe the synthesis/removal, nuclei and projections and receptors tied to catecholamines and serotonin

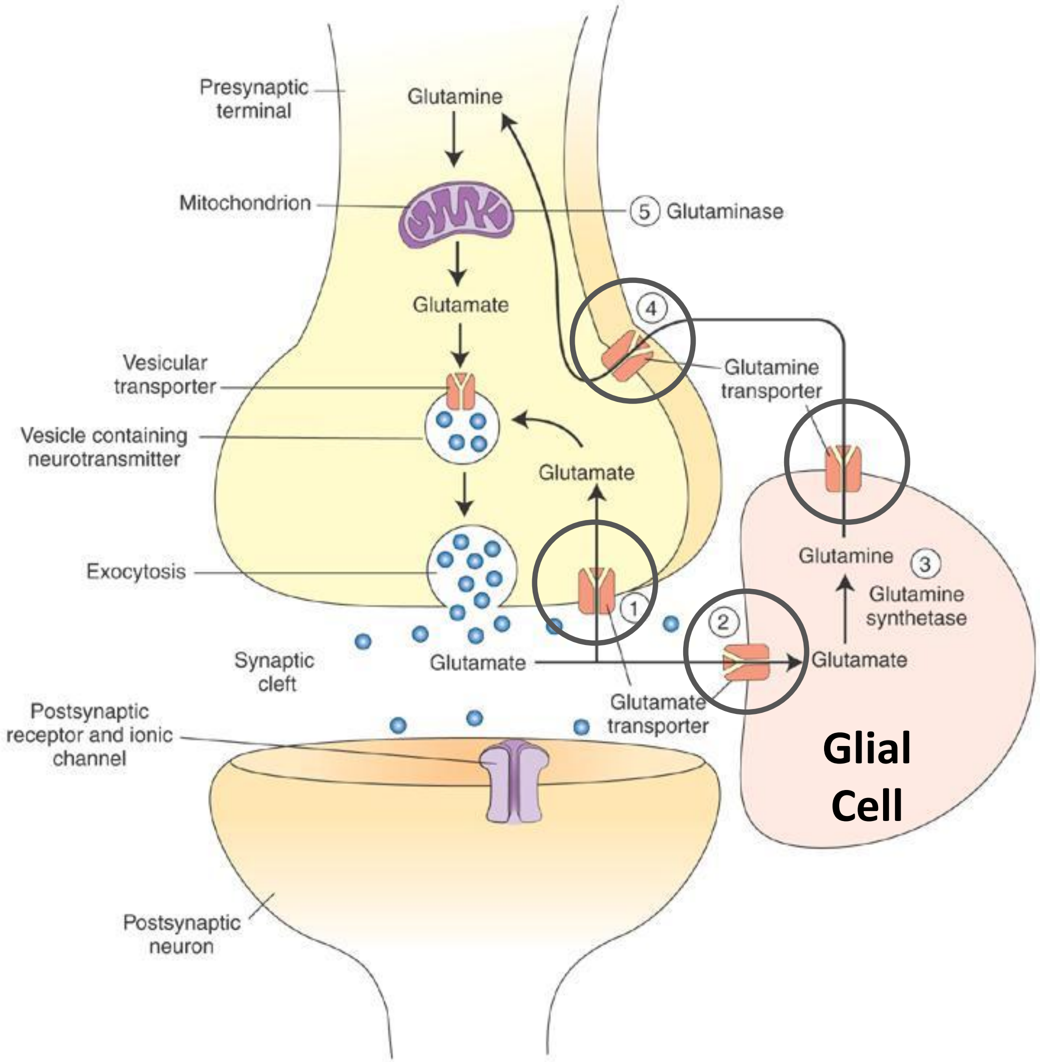
Learning Objectives

- **SOM.MK.I.BPM2.1.NB.1.NSCI.0111** Describe the nuclei and projections and receptors tied to acetylcholine

- **SOM.MK.I.BPM2.1.NB.1.NSCI.0112** Describe the nuclei and projections and receptors tied to amino acid transmitters

- **SOM.MK.I.BPM2.1.NB.1.NSCI.0113** Describe the nuclei and projections and receptors tied to catecholamines and serotonin

Neurotransmitter Systems: Glutamate



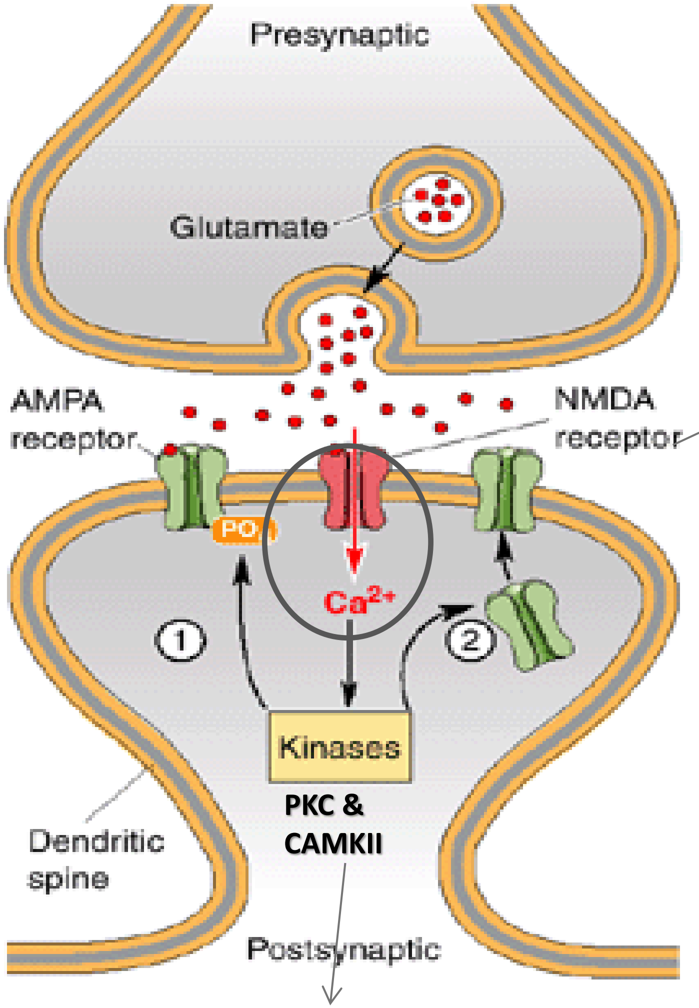
Nuclei and Projections

Ubiquitous throughout CNS

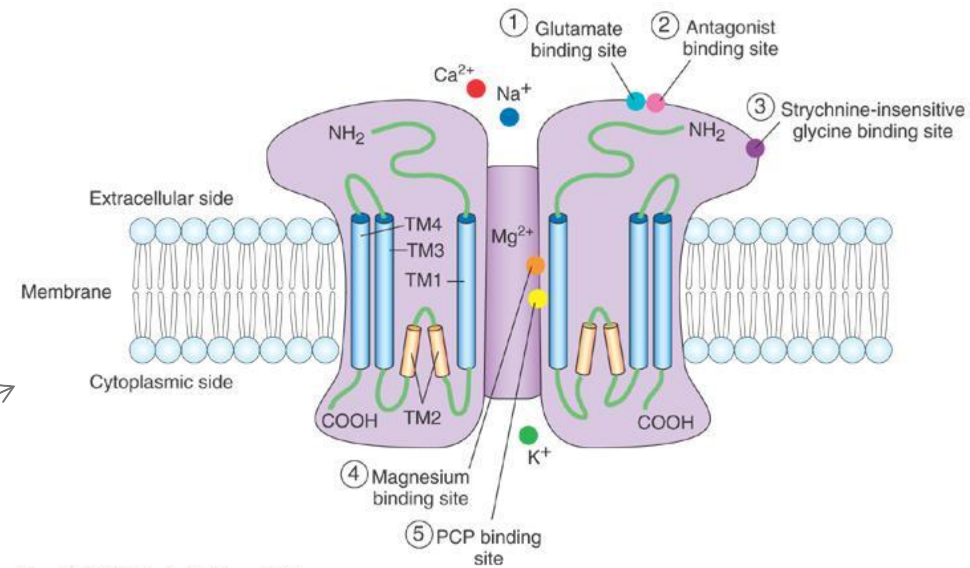
Fig 7.7, Siegel & Sapru.

Please refer to DLA

Glutamate receptors



Promotes spine formation in LTP



- Ionotropic glutamate receptors
  - AMPA and Kainate receptors induce Na/K flux after glutamate binding
  - EPSP summation unblocks the NMDA channel pore → Ca flux
- Metabotropic glutamate receptors
  - Group 1 mGLUR – (e) 1,5
  - Group 2 mGLUR – (i) 2,3
  - Group 3 mGLUR – (i) 4,6-8

Fig 25.5, Siegel & Sapru.

Glutamate: Pathology and Therapeutics

- Seizures
  - Heightened release of glutamate through hyper-synchronization of neuronal populations
  - NMDA antagonists can be used to suppress seizures
    - Reduced influx of calcium
- Weakness
  - **Reduced release** of glutamate into bulbar and spinal motor nuclei
    - As seen in Upper Motor Neuron Syndrome

Gamma-Aminobutyric Acid (GABA)

Nuclei and projections:

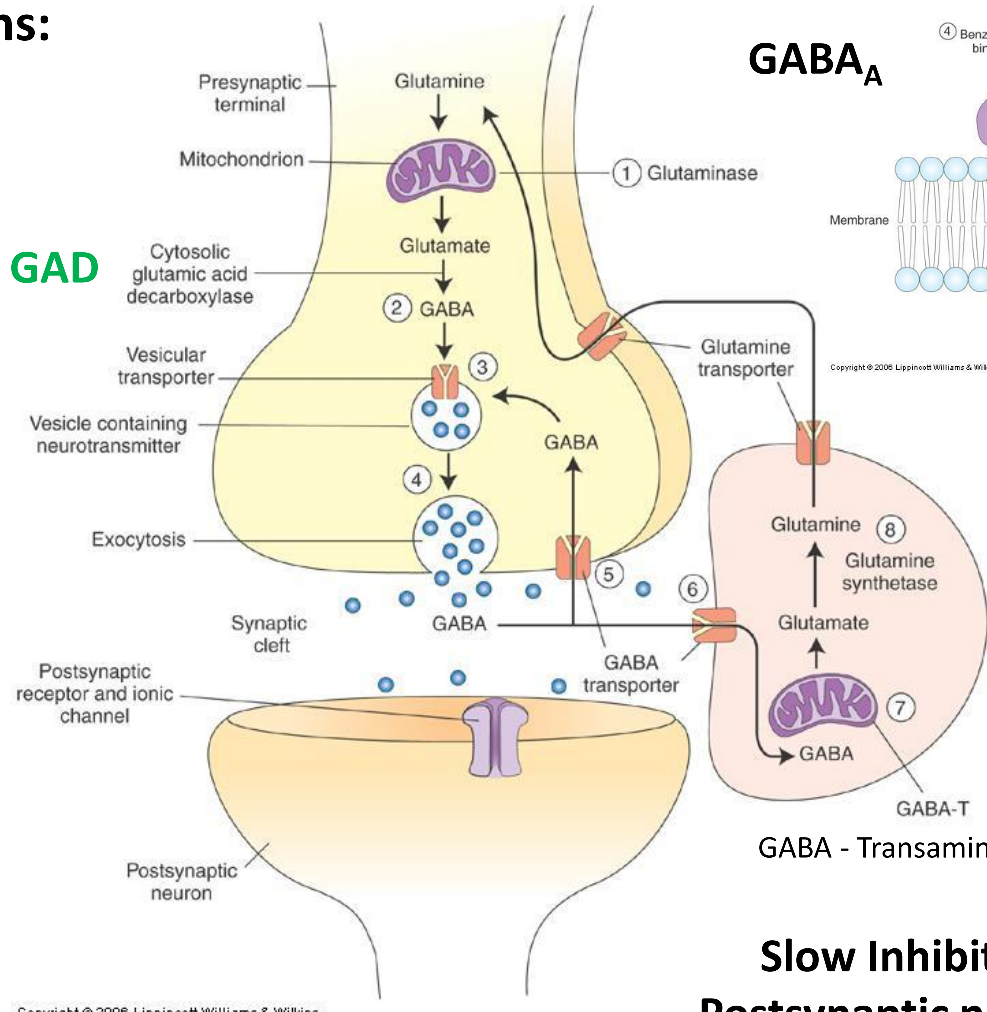
GABAergic interneurons are ubiquitous

GABAergic pathways:

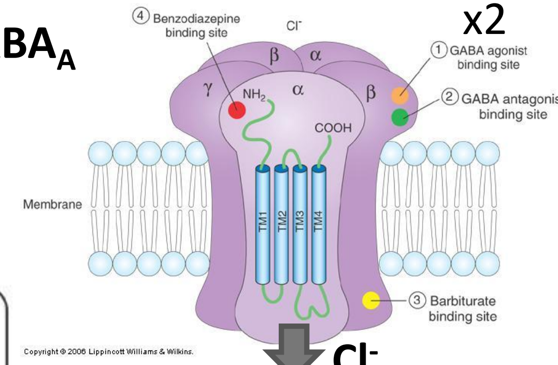
- Striatum -> substantia nigra
- Substantia nigra -> superior colliculus and thalamus
- Medial vestibular nuclei --> spinal cord
- Cerebellar cortex -> deep cerebellar nuclei

Please refer to DLA

GAD



GABA$_A$



Cl$^-$

Anxiolytic Drugs

- Prolong Opening

GABA$_B$

Metabotropic

2nd Msng

K$^+$

Slow Inhibition Postsynaptic neurons

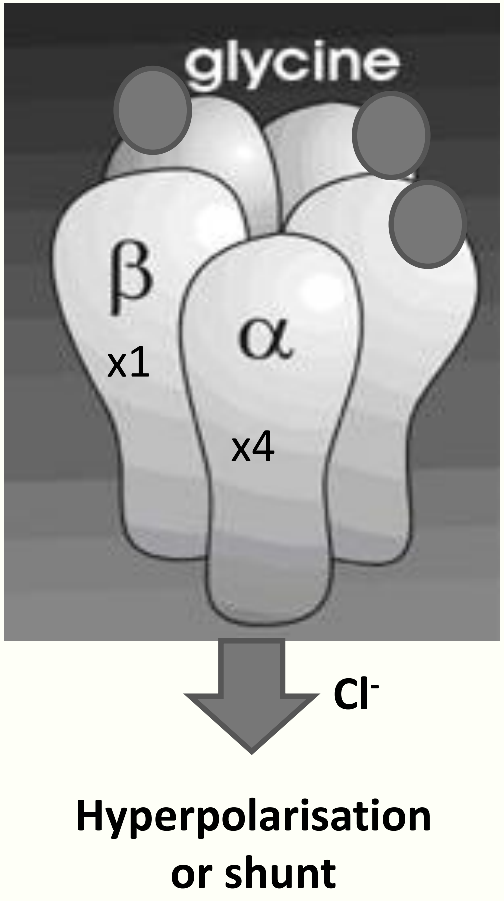
GABA - Transaminase

Figure from Siegel & Sapru.

GABA: Pathology and Therapeutics

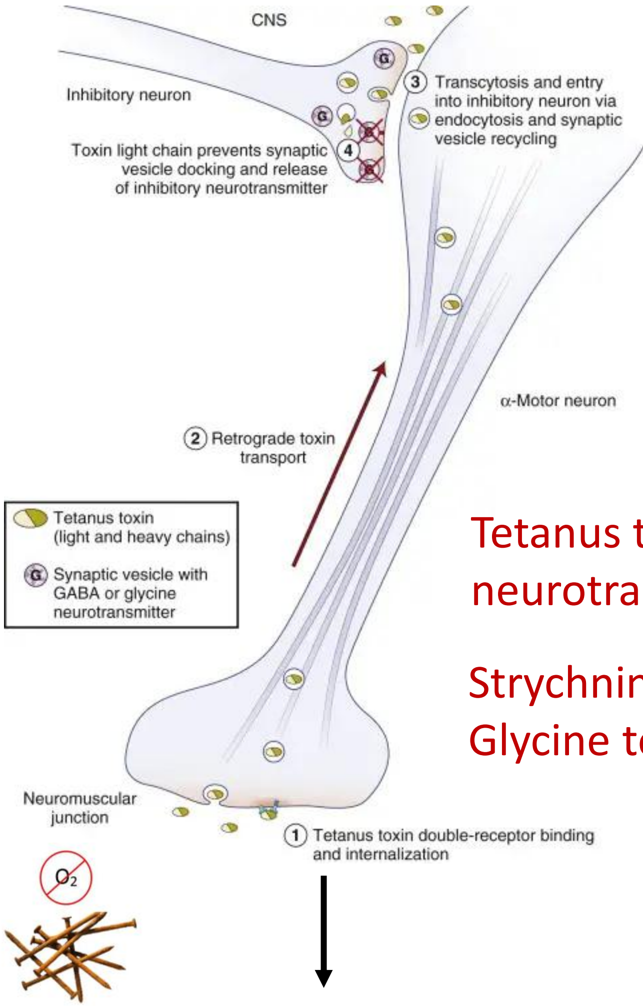
- Seizures
  - Heightened release of glutamate through hyper-synchronization of neuronal populations
  - Treatment
    - **GABA agonists** can **inhibit** hyper-excitable cells
- Anxiety
  - Mediated through altered excitability of limbic cells
  - Treatment
    - Benzodiazepines **augment GABAergic transmission**
      - **Reducing excitability**

Glycine

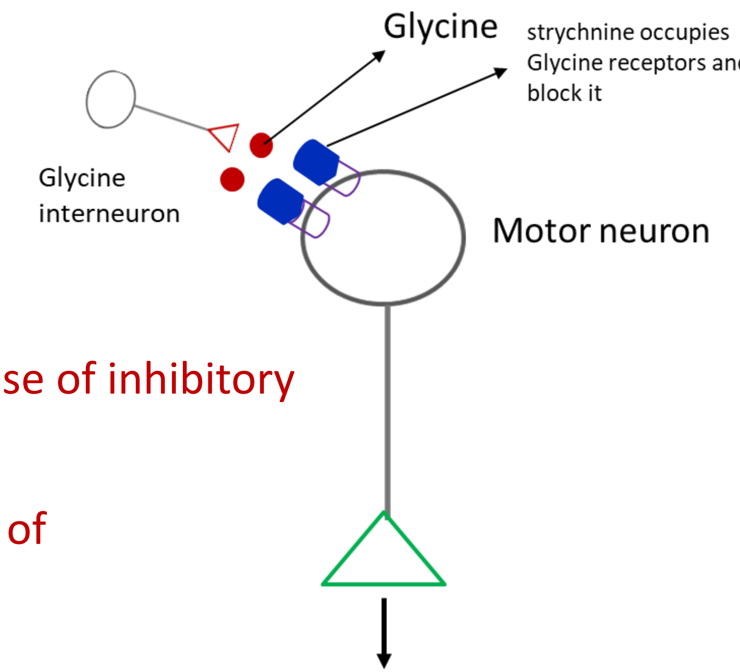


- Nuclei and projections
  - Neurons tend to be **small local** inhibitory Interneurons
    - Common in spinal and bulbar motor nuclei
  - Only one receptor type
    - Ionotropic with a **Cl^- channel**
      - Blocked by **strychnine**

Glycine: Pathology and Therapeutics
Tetanus Vs. Strychnine poison



Muscle contraction



Muscle contraction

Tetanus toxin: Prevents release of inhibitory neurotransmitter(Glycine)

Strychnine: Prevents binding of Glycine to receptors

Lower motor neurons are disinhibited in both the poisoning leading to strong muscle contraction

SOM.MK.I.BPM2.1.NB.1.NSCI.0112

https://microbeonline.com/clostridium-tetani-properties-pathogenesis-diagnosis/

Glycine: Pathology and Therapeutics

- Tetanus
  - Toxic protease **suppresses release** of glycine from bulbar and spinal interneurons
    - Lower motor neurons are **disinhibited**
  - Deploying muscle-relaxing strategies directed at central or peripheral targets
- Strychnine poisoning
  - The alkaloid **blocks glycine** receptor on lower motor neurons, reducing chloride influx
    - Lower motor neurons are **disinhibited**

Lecture Objectives

SOM.MK.I.BPM2.1.NB.1.NSCI.0111	Describe the synthesis/removal, nuclei and projections and receptors tied to acetylcholine
SOM.MK.I.BPM2.1.NB.1.NSCI.0112	Describe the synthesis/removal, nuclei and projections and receptors tied to amino acid transmitters
SOM.MK.I.BPM2.1.NB.1.NSCI.0113	Describe the synthesis/removal, nuclei and projections and receptors tied to catecholamines and serotonin

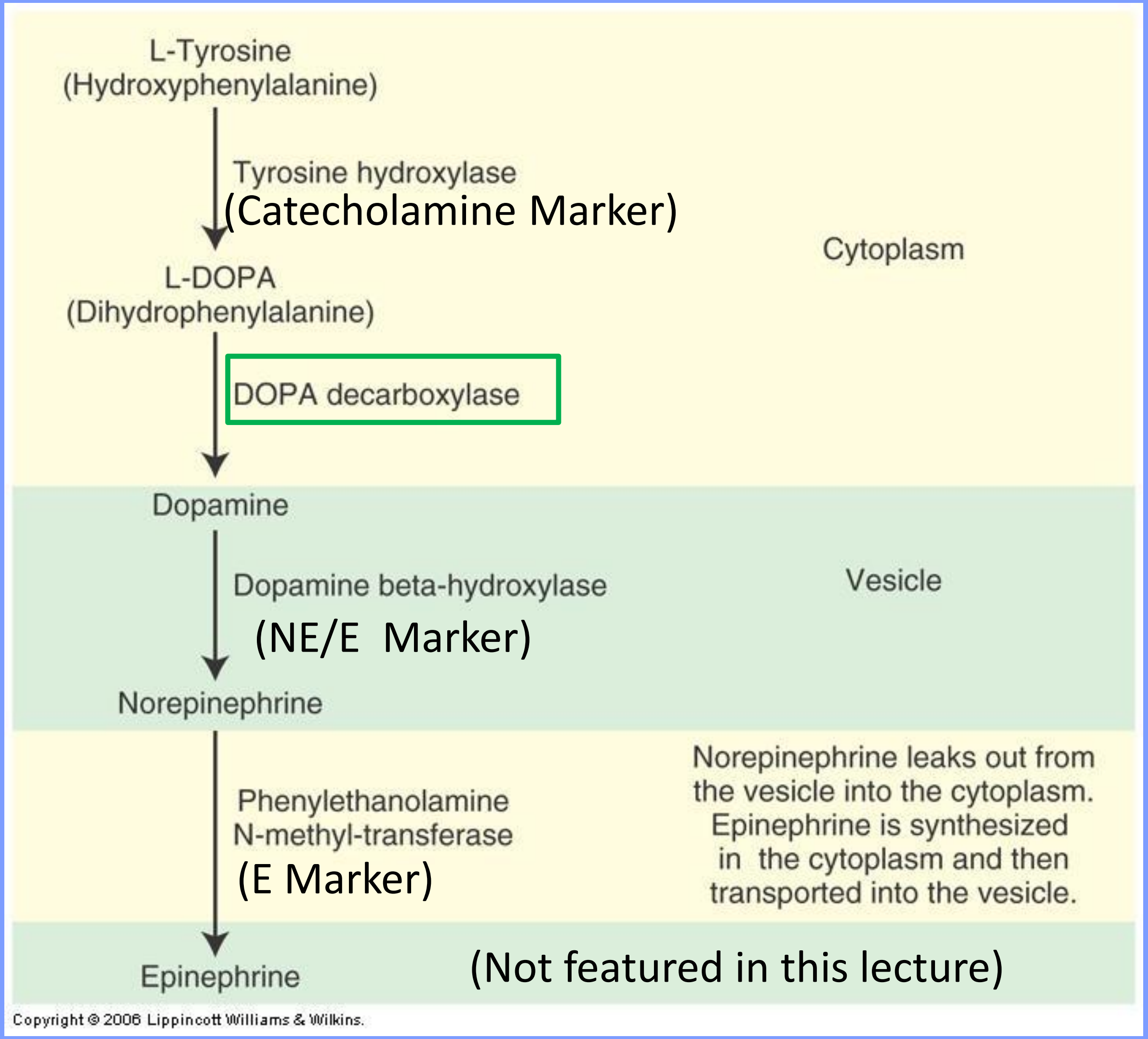
Learning Objectives

- **SOM.MK.I.BPM2.1.NB.1.NSCI.0111** Describe the nuclei and projections and receptors tied to acetylcholine

- **SOM.MK.I.BPM2.1.NB.1.NSCI.0112** Describe the nuclei and projections and receptors tied to amino acid transmitters

- **SOM.MK.I.BPM2.1.NB.1.NSCI.0113** Describe the nuclei and projections and receptors tied to catecholamines and serotonin

Catecholamines

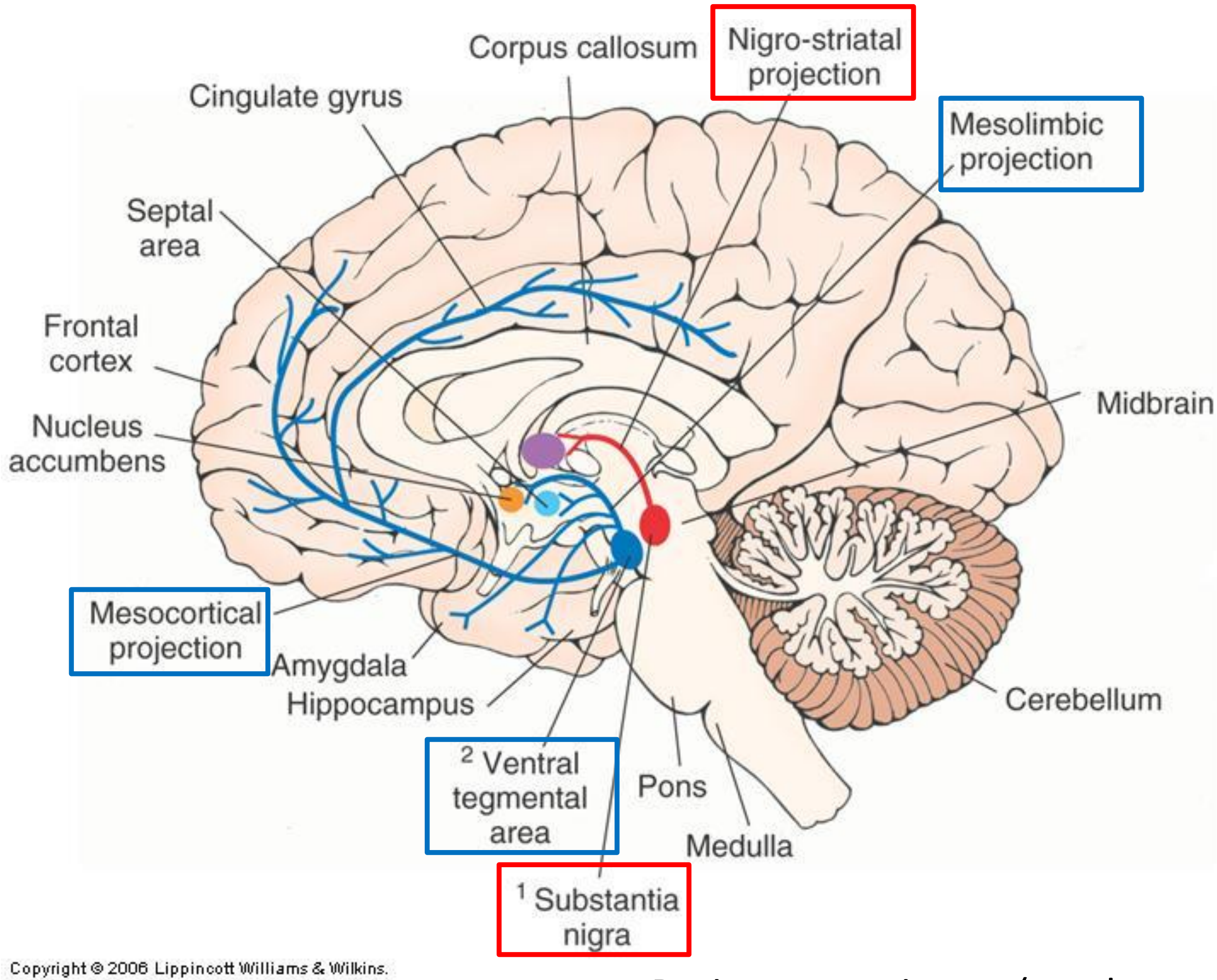


If a cell is DBH positive but PNMT negative, then it synthesizes NE but not E

Please refer to DLA

Figure from Siegel & Sapru.

Dopamine: CNS pathways & receptors



Projects to striatum (caudate nucleus and putamen)

All Metabotropic receptors:

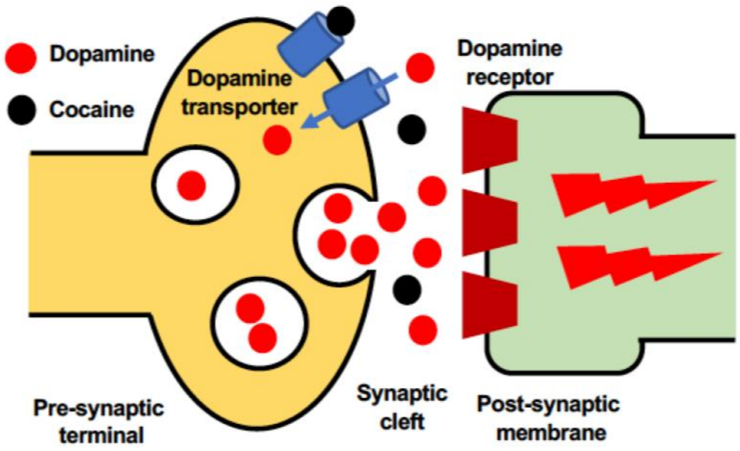
D_1-like – increases cAMP (D_1/D_5)
D_2-like – Decreases cAMP (D_2-D_4)

Reward, mood and movement

Fig 7.11, Siegel & Sapru.

Dopamine: Pathology and Therapeutics

The Neurotoxicity of Cocaine

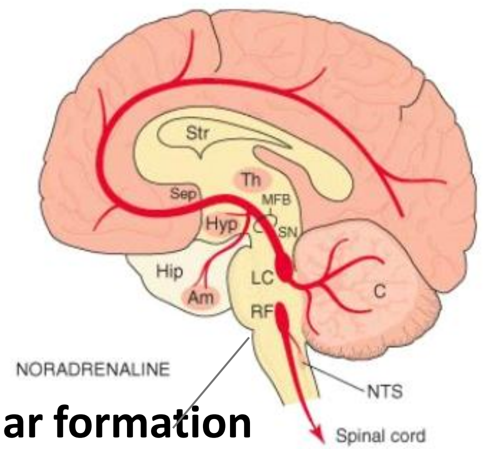


- Movement Disorders
  - Parkinsonism
    - Decreased movement
    - Treatment
      - Dopamine agonists
- Psychoses
  - Hallucinations and delusions
  - Treatment
      - Dopamine antagonists (major tranquilizers)

https://doi.org/10.3390/ijms23105418

Catecholamine Neurotransmitters: Norepinephrine (noradrenaline)

Central projections



Reticular formation

LC; Locus Coeruleus – diencephalon, limbic system (emotional centre), cerebral cortex (sensory pathways- micturition, cerebellum and spinal cord (connection not shown)

Please refer to DLA

Synthesis , Release and degradation

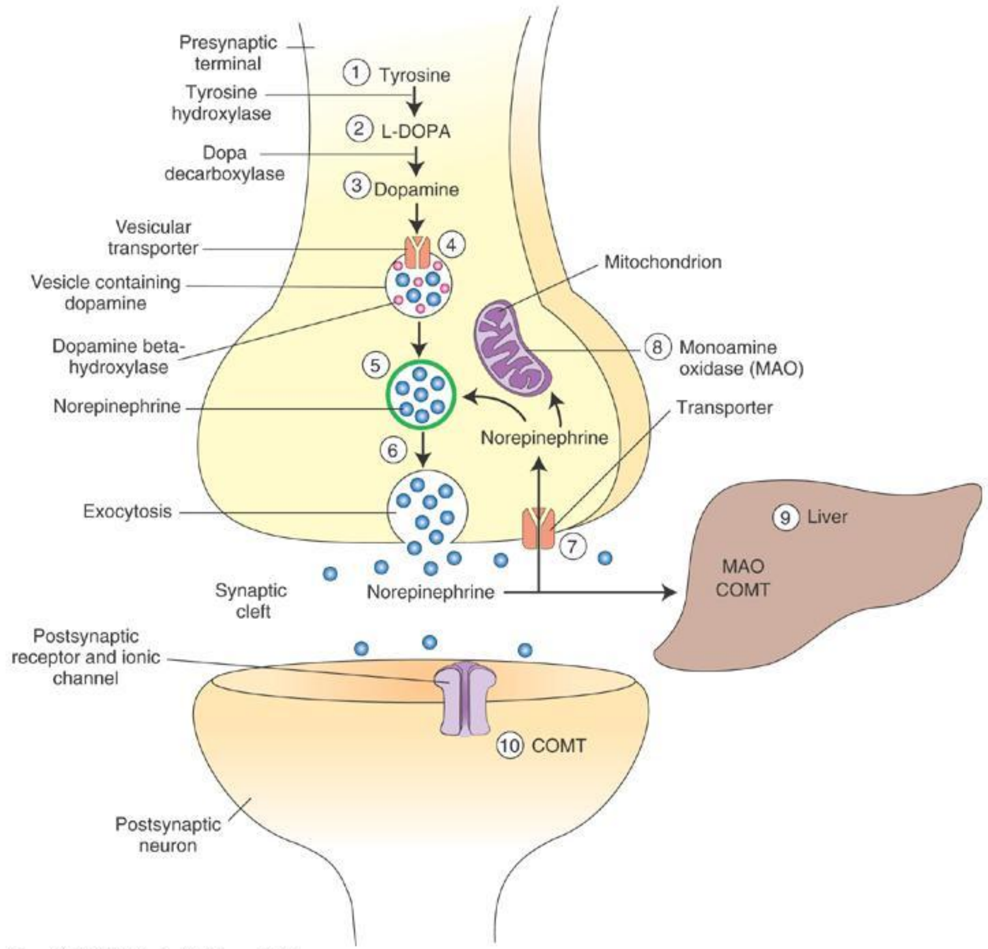
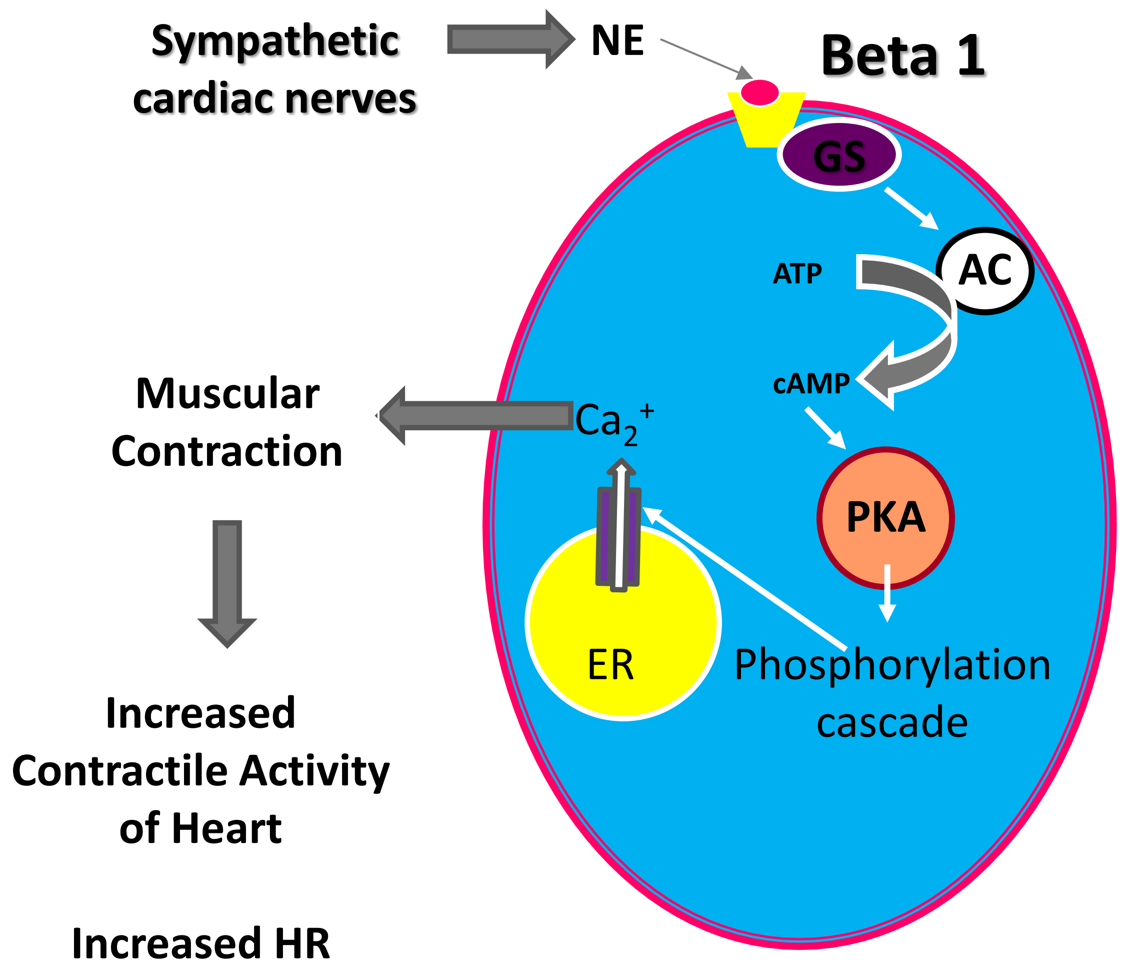


Fig 7.12, Siegel & Sapru.

Norepinephrine Receptors

- Metabotropic
- $Alpha_1/Beta_1/Beta_2$
  - Excitatory
- $Alpha_2$
  - Inhibitory
- Postganglionic sympathetic neurons excite cardiac muscle

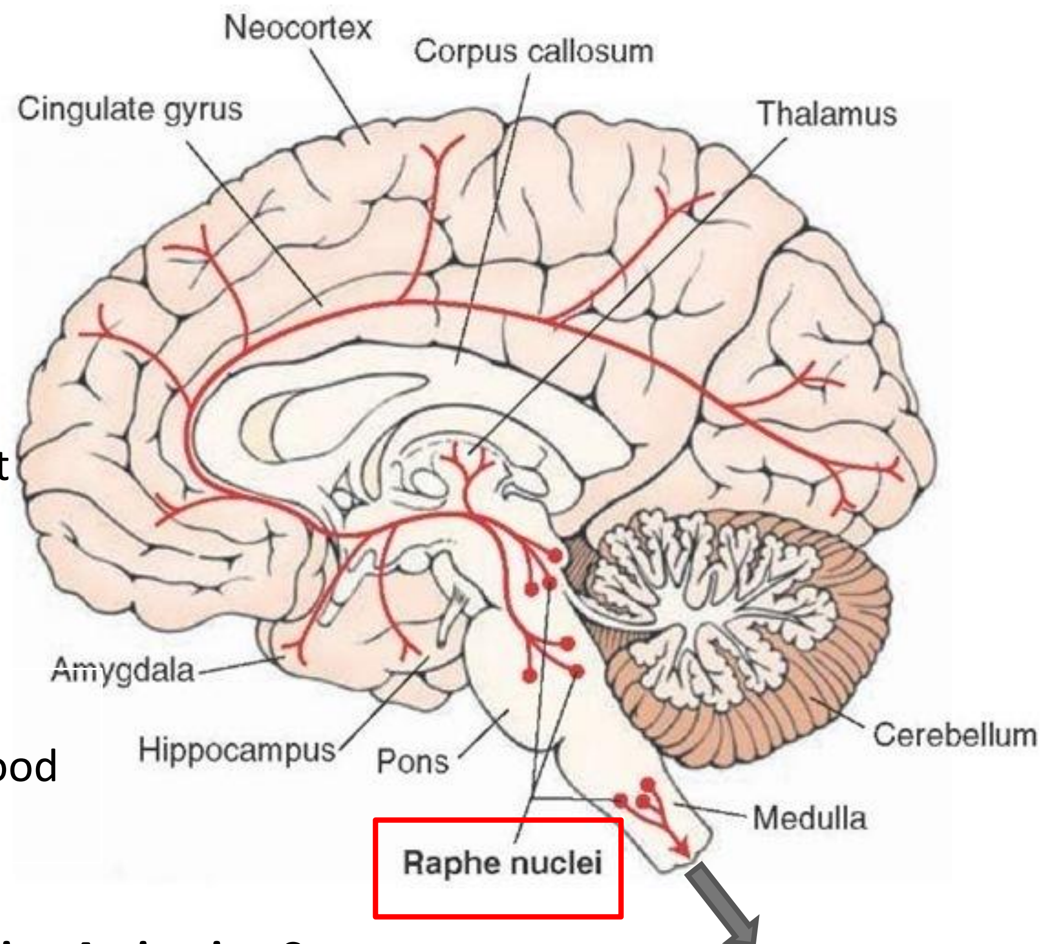


Norepinephrine: Pathology and Therapeutics

- Depression and Pain
  - Treatment
    - **Norepinephrine agonists** (re-uptake or MAO inhibitors)
      - Activates spinal and bulbar opioidergic neurons
        - Suppresses release of substance P onto spinal dorsal horn and spinal trigeminal nucleus
- Sympathetic insufficiency
  - Parkinson’s disease
    - Orthostatic hypotension
      - Loss of peripheral sympathetic neurons
      - Treatment
        - **Norepinephrine agonists**

Serotonin CNS pathways: Mood & Arousal

Noradrenergic pathways from Locus Coeruleus broadly similar – see previous slide



Frontal Cortex
- Social Context
- Appropriate behaviours

Limbic system
- Emotions/mood
- Anxiety/Fear

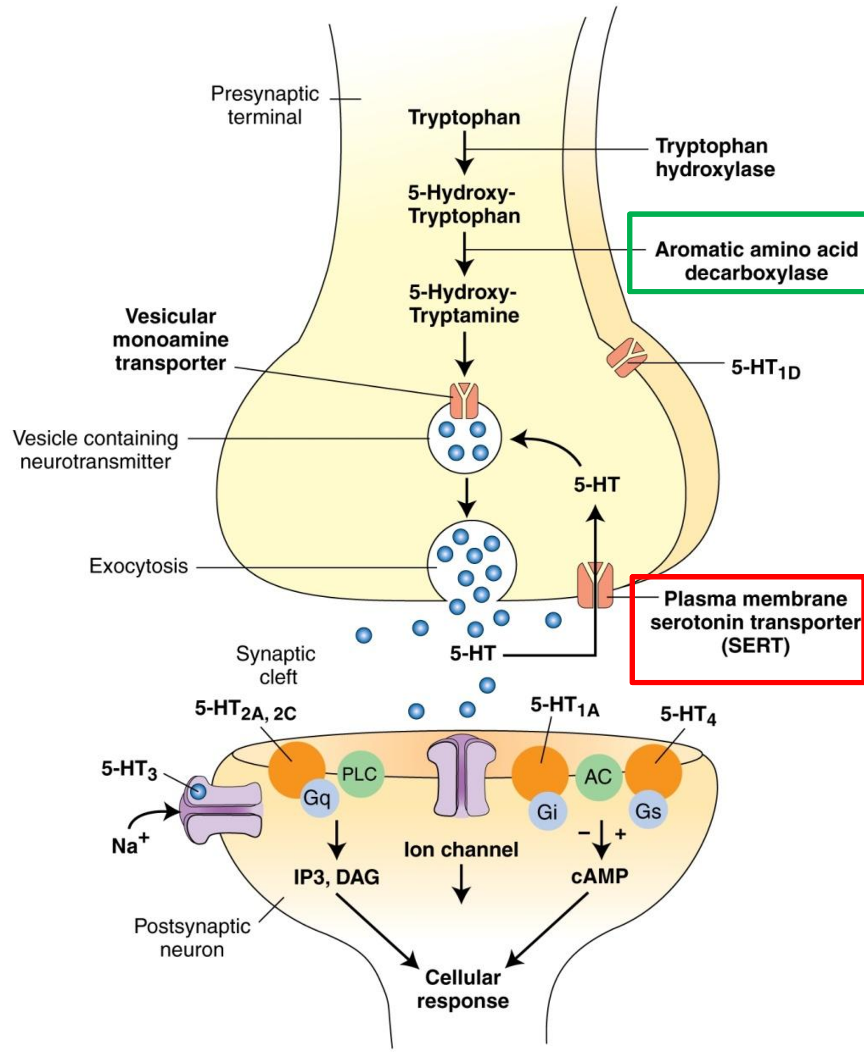
Ascending Activating System
- Arousal/Wakefulness

To spinal Cord, modulates nociception and sexual reflexes

SOM.MK.I.BPM2.1.NB.1.NSCI.0113
Fig 7.15, Siegel & Sapru.

Serotonin receptors

- Seven families 5-HT_1 -5-HT_7 and further subtypes, e.g. 5-HT_{1A}
- **All metabotropic** with **exception of 5-HT_3**
  - 5-HT_1, 5-HT_5: inhibitory
  - 5-HT_2: excitatory
  - **5-HT_3: excitatory ionotropic** (cation-permeable)
  - 5-HT_4, 5-HT_6, 5-HT_7: excitatory
- Complex interactions of different 5-HT receptor isoforms can co-exist in the same neuron, e.g. spinal neurons governing micturition



Please refer to DLA

Figure from Siegel & Sapru.

Serotonin: Pathology and Therapeutics

- Depression and Pain
  - Treatment
    - Serotonin agonists (re-uptake or MAO inhibitors)
      - Activates spinal and bulbar opioidergic neurons
        - Suppresses release of substance P onto spinal dorsal horn and spinal trigeminal nucleus

End of Neurotransmitter Pathways and Roles

Thank you

For questions and clarifications

please email

nleton@sgu.edu